

Ethics, Privacy and Security

Ethical considerations

The project uses aggregated pharmaceutical dispensing data to investigate patterns involving donepezil hydrochloride, rivastigmine and other widely distributed medications across New Zealand health districts. Because the data is aggregated geographically and does not capture all factors that may influence medication dispensing, relationships observed in the analysis cannot be assumed to represent individual behaviour or direct causal relationships.

In particular, an observed relationship between socioeconomic deprivation and medication dispensing should not be interpreted as evidence that deprivation directly causes differences in medication use. Differences between districts may also reflect factors that are not included in the analysis, such as access to healthcare, prescribing practices, demographic composition, population size, or differences in the prevalence and diagnosis of particular conditions. For this reason, the analysis describes associations and trends rather than causal relationships.

Care is also required when discussing areas with relatively high levels of deprivation. Results should be presented in a way that avoids stigmatising particular districts or populations. A higher or lower dispensing rate does not automatically indicate better or worse healthcare because observed dispensing patterns may reflect several demographic, clinical and structural factors.

Māori data sovereignty is also relevant when analysing health information in Aotearoa New Zealand. Although this project uses aggregated data rather than identifiable Māori health records, analyses of geographic and socioeconomic health patterns may still contribute to conclusions about Māori communities. Results involving Māori populations should therefore be interpreted and communicated respectfully and within their wider social and historical context. With access to more detailed population-level or ethnicity-specific health information, the project would require greater consideration of Māori governance over the collection, interpretation, storage and use of Māori data.

Privacy considerations

The pharmaceutical dataset used in the project is aggregated at health-district level rather than individual-patient level. Individual-level medication records, names, NHI numbers and other direct patient identifiers were therefore neither required nor used in the analysis.

Small counts in the pharmaceutical dataset are represented as "<6" to reduce the risk of identifying patients in groups with very few observations. In our analysis, these suppressed observations were identified and retained using suppression indicators. Where numerical values were required for analysis, suppressed counts were approximated as 3, the midpoint of the possible range from 1 to 5. This allows suppressed observations to remain in the analysis without claiming that their true confidential values are known.

This approximation introduces uncertainty, particularly for medications or districts with relatively small dispensing counts. Results affected by these observations should therefore be interpreted cautiously, and the imputed value of 3 should not be presented as the actual underlying count.

Suppression also does not completely eliminate disclosure risk. In some circumstances, suppressed values could potentially be inferred by comparing totals, subtotals, years, medication categories or other overlapping published aggregates. For example, if a total and most of its component counts were available, the remaining suppressed value could potentially be estimated through subtraction. We did not attempt to reverse-engineer suppressed observations; instead, suppressed values were consistently treated as unknown values within the range of 1 to 5 and approximated only where required for analysis. This illustrates why privacy must be considered across combinations of outputs rather than only within individual tables.

Security considerations

For this project, we worked only with the datasets required for the analysis. The pharmaceutical data was already aggregated and did not contain names, NHI numbers or other direct patient identifiers. The original pharmaceutical dataset was loaded without being overwritten, while cleaned and transformed versions were created as separate data objects and derived output files. This helps preserve the original source while allowing the analytical transformations to remain reproducible.

The analysis code also explicitly records important processing decisions, including the identification and treatment of suppressed values, the restriction of the main analysis to total dispensings, the harmonisation of district geographies and the creation of population-adjusted dispensing rates. Keeping these transformations in code allows them to be reviewed and reproduced rather than being performed manually.

We also avoided intentionally reconstructing suppressed values when producing analytical outputs. However, because suppressed values may sometimes be inferred by combining multiple aggregates, tables and visualisations should be reviewed before publication to ensure that combinations of reported results do not inadvertently reveal small confidential counts.

With greater resources, and particularly if individual-level or more detailed health data were used, stronger security controls would be required. These could include restricted user access, secure research environments, encrypted storage and transfer, formal approval processes, audit logs and systematic disclosure-risk assessment before releasing results. More detailed Māori health data would also require stronger consideration of Māori data sovereignty and appropriate Māori governance over access, storage, interpretation and publication.